SIAMESE NETWORKS FOR FACE VERIFICATION ON CELEBA: A COMPARATIVE STUDY OF TRIPLET AND CONTRASTIVE LEARNING

A PROJECT REPORT

Department of Computer Science
Aarhus University

Department of Computer Science
Aarhus University

Department of Computer Science
Aarhus University

December 5, 2025

ABSTRACT

Face verification requires learning embedding spaces where images of the same person remain close despite changes in pose, lighting, and appearance. This project addresses this challenge using two Siamese-network approaches on the CelebA dataset. First, a MobileNetV2 backbone trained with Triplet Loss and strong augmentation and regularisation achieves an AUC of 0.8009 and a verification accuracy of 72.12%. Second, a ResNet18-based Siamese model trained with Contrastive Loss, combined with augmentation and partial fine-tuning, reaches an AUC of 0.91 and an accuracy of 82.95%. Beyond these quantitative gains, a qualitative study using t-SNE embeddings, misclassified pair analysis, and Grad-CAM visualisations provides insight into model behaviour and remaining failure modes. Together, these results offer a unified view of lightweight Siamese architectures and the factors shaping their performance in face verification.

Keywords Face verification · Siamese networks · Metric learning · Contrastive loss

1 Introduction

Automated face recognition has emerged as one of the most significant applications of computer vision, forming the foundation for technologies ranging from biometric security systems to social media tagging and personalized user experiences. While large-scale classification networks can achieve impressive accuracy, they remain limited in verification scenarios, where the goal is not to identify a person among many classes, but to determine whether two images belong to the same identity. This shift from *closed-set* classification to *open-set* verification requires learning an embedding space that preserves meaningful facial similarity under changes in pose, illumination, age, and expression.

Siamese networks offer a natural solution to this challenge. By processing two images through identical CNN backbones and optimising a distance-based objective, they enable the model to learn compact intra-class clusters and well-separated inter-class boundaries. The choice of backbone architecture, loss function, and training strategy plays a central role in shaping the resulting embedding space.

In this project, we explore face verification on the CelebA dataset through two parallel Siamese-network directions. The first employs MobileNetV2 and Triplet Loss, focusing on efficient feature extraction and strong regularisation. The second develops a ResNet18-based Siamese pipeline trained with Contrastive Loss, where we systematically study the effect of data augmentation, partial fine-tuning, and regularisation. Rather than treating these as competing approaches, we view them as *complementary perspectives* on the same problem: lightweight architectures paired with metric-learning objectives.

Our objectives are threefold:

- To implement reproducible Siamese verification pipelines using lightweight CNN backbones.

- To compare Triplet- and Contrastive-Loss training under different augmentation, tuning, and regularisation scenarios.
- To evaluate model behavior through quantitative metrics for all approaches and qualitative analysis for the top-performing model, using tools such as t-SNE embeddings, misclassified-pair inspection, and Grad-CAM visualisations.

Together, these investigations aim to provide a clear understanding of how architectural choices and training strategies influence the performance and robustness of Siamese networks for face verification.

2 Related Work

Early work on verification with neural networks introduced the Siamese architecture as a way to learn similarity directly from paired data. Bromley et al. proposed a "Siamese" time-delay neural network for online signature verification, where twin subnetworks process two inputs and a shared distance head predicts whether they match [Bromley et al., 1993]. This idea was later adapted to images in the context of one-shot classification: Koch et al. showed that Siamese convolutional networks can recognise novel classes from a single example by learning a generic similarity metric over images [Koch et al., 2015].

In face recognition, large-scale systems such as DeepFace and related models first demonstrated that deep CNNs can approach or even surpass human-level performance on benchmarks like LFW by treating the problem as closed-set classification over thousands of identities [Taigman et al., 2014]. These methods typically learn a high-capacity backbone and a softmax classifier, and only derive a verification embedding indirectly from intermediate features.

Subsequent work shifted towards metric-learning objectives that more directly optimise a face embedding space. FaceNet popularised the use of Triplet Loss to learn a Euclidean embedding where distances correspond to face similarity, combined with online hard/semi-hard triplet mining and large-scale training on datasets such as MS-Celeb-1M [Schroff et al., 2015]. More recent approaches like ArcFace replace Triplet Loss with margin-based variants of the softmax classifier, introducing additive angular margins to enforce stronger class separation on a hypersphere [Deng et al., 2019]. These methods define the current state of the art on standard verification benchmarks, but typically rely on very deep backbones (e.g. ResNet-100) and millions of training identities.

Siamese architectures have also been applied directly to face verification under specific challenges using pairwise losses. For example, Bianco et al. studied large age-gap verification by fine-tuning a DCNN in a Siamese configuration with Contrastive Loss, showing that incorporating external features through feature injection can improve robustness across large age differences [Bianco, 2017]. Other works extend Siamese ideas to related tasks such as identity-preserving super-resolution or liveness detection, again using distance-based objectives to maintain identity consistency.

In parallel, there has been a strong push towards lightweight backbones that make face recognition feasible on mobile or embedded devices. MobileNetV2 introduced an architecture based on inverted residual blocks and linear bottlenecks, greatly reducing parameter count and FLOPs while preserving competitive accuracy [Sandler et al., 2018]. Building on this, a number of lightweight face recognition models such as MobileFaceNet and its variants adapt mobile-style backbones to face embedding tasks, showing that compact networks can still achieve strong verification performance when paired with appropriate losses and training strategies [Chen et al., 2022].

Our work situates itself between these lines of research. On one hand, we adopt metric-learning objectives (Triplet and Contrastive Loss) and Siamese architectures that directly optimise an embedding space for face verification, inspired by FaceNet and contrastive Siamese models. On the other hand, we focus on lightweight backbones and moderate-scale data, using MobileNetV2 and ResNet18 rather than very deep architectures. Unlike leaderboard-oriented studies that emphasise a single best model on large proprietary datasets, our goal is to compare two practical Siamese configurations under controlled conditions on CelebA, and to analyse how design choices augmentation strength, partial fine-tuning, and regularisation affect both quantitative performance and qualitative behaviour of the learned embeddings.

3 Methods

This section presents the dataset, similarity-construction strategy, the two Siamese-network architectures explored, and the distinct training pipelines used for the TensorFlow and PyTorch tracks.

3.1 Dataset

All experiments use a CelebA-like subset [1] containing cropped face images, each annotated with an identity label. The dataset retains substantial variation in pose, illumination, and facial appearance, making the verification problem non-trivial. Images are resized to 109×89 pixels and normalised to $[0, 1]$. We adopt an identity-level train/test split (80%/20%) to ensure that evaluation is strictly open-set.

3.2 Pair and Triplet Construction

To support both Siamese-loss formulations, we construct two forms of supervision:

- **Pairs (Contrastive Loss):** We sample balanced positive and negative pairs from identity labels. Positive pairs contain two images of the same identity; negative pairs are sampled across identities.
- **Triplets (Triplet Loss):** For MobileNetV2, we construct anchor–positive–negative triplets using random identity sampling. Triplets are generated without online mining so that both pipelines operate under comparable conditions.

Both formulations aim to shape an embedding space where distance reflects identity similarity.

3.3 Backbones and Embedding Heads

We examine two lightweight Siamese architectures, chosen for complementary reasons: MobileNetV2 for efficiency under Triplet Loss, and ResNet18 for controlled ablation under Contrastive Loss.

MobileNetV2 + Triplet Loss (TensorFlow): The MobileNetV2 branch uses an ImageNet-pretrained backbone with the classification layers removed. The feature tensor is passed through an embedding head comprising:

- Dense layer (256 units, ReLU) with L2 regularisation,
- Batch Normalization,
- Dropout (0.5),
- Dense layer producing 128-dimensional embeddings,
- L2 normalisation to stabilise Triplet-Loss geometry.

This architecture emphasises computational efficiency and strong regularisation, reflecting the constraints of triplet-based training.

ResNet18 + Contrastive Loss (PyTorch): In parallel, we develop a ResNet18-based Siamese network with:

- A shared ResNet18 backbone between branches,
- A linear projection to a 128-dimensional embedding,
- L2 normalisation for scale consistency,
- Two training modes: fully frozen backbone, and a partially fine-tuned variant unfreezing `layer4`,
- Optional dropout and weight decay evaluated during ablations, though not part of the final configuration.

This design enables controlled study of how fine-tuning depth and augmentation affect pairwise distances in Euclidean space. A unified overview of both architectures is provided in Figure 1.

3.4 Training Strategy

Optimisers and Schedulers: Both pipelines use the Adam optimizer with an initial learning rate of 10^{-4}. The TensorFlow branch incorporates:

- `ReduceLROnPlateau` (factor 0.2, patience 2),
- `EarlyStopping` (patience 5), saving the best-performing model.

[1]Dataset link: `https://drive.google.com/drive/folders/ ?usp=share_link`

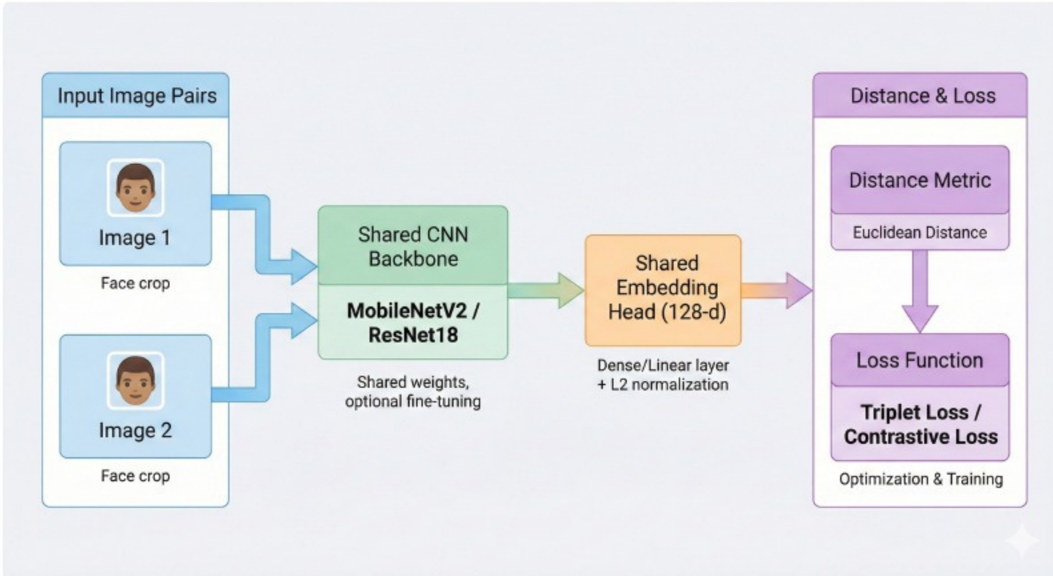


Figure 1: Overview of the two Siamese configurations explored in this project. Both tracks share the same twin-branch design but differ in backbone choice and loss formulation.

The PyTorch branch mirrors this behaviour conceptually: All experiments run for 10 epochs, the last and checkpoint with the best validation metrics are kept.

Epoch Count and Model Selection. Triplet-based training converges more slowly; MobileNetV2 models typically trained for 15–20 epochs. The ResNet18 Contrastive-Loss models often reached optimal separation within 5–10 epochs. In both tracks, final performance is reported using the best validation-epoch model rather than the final epoch.

3.5 Data Augmentation and Regularisation

Both experimental tracks relied on data augmentation to increase invariance to common facial variations, but the specific augmentations differed according to the implementation framework and training objective.

MobileNetV2 (TensorFlow): This branch applied a lightweight augmentation block implemented via `tf.keras.Sequential`, consisting of horizontal flips, small rotations, mild zoom, and low-intensity contrast jitter: `RandomFlip`, `RandomRotation(0.1)`, `RandomZoom(0.1)`, and `RandomContrast(0.1)`. No occlusion-based augmentation was used. The aim was to stabilise triplet training without substantially altering local facial structure, given the relatively low input resolution (109×89).

ResNet18 (PyTorch): This branch employed three augmentation presets, implemented via `build_siamese_transform` ():

- **none**: resize and normalisation only (baseline).
- **light**: horizontal flip, small rotation (10°), and mild colour jitter (brightness/contrast/saturation 0.2), encouraging robustness to pose and illumination changes.
- **strong**: all light augmentations plus **RandomErasing** (p=0.5), introducing small occlusion patches (2–20% of the image).

Occlusions were kept intentionally mild: on 109×89 inputs, larger masking regions risk removing entire facial components, which can destabilise contrastive learning. The strong preset therefore targets moderate robustness gains without over-occluding identity-critical features.

Regularisation: Both tracks employed classical regularisation mechanisms to prevent overfitting on the moderate-sized CelebA subset. Dropout rates between 0.3–0.5 and weight decay (10^{-4}) were evaluated, with the final ResNet18 configuration using dropout selectively. Partial fine-tuning (unfreezing `layer4`) served as an additional regulariser by adapting high-level features while preserving earlier layers. As seen in the Results, insufficient regularisation led to rapid memorisation, whereas overly strong regularisation (e.g. excessive dropout in the R3 configuration) compressed the embedding space and slightly reduced AUC.

3.6 Implementation Details

The two tracks differ in tooling and engineering focus:

- **MobileNetV2 (TensorFlow):** This branch focused on making triplet-based training on $N = 10{,}000$ triplets per epoch computationally practical by progressively optimising throughput. Starting from a CPU baseline (one epoch $\approx$ 1313s), moving to an NVIDIA T4 GPU cut epoch time to $\sim$ 91.5s, and an asynchronous `tf.data` pipeline reduced this further to $\sim$ 75s by eliminating GPU idle time. Enabling mixed-precision training (`mixed_float16`) brought epoch time down to $\sim$ 70s and improved validation accuracy (71.70% to 75.05%), while temporarily freezing the first 80 layers lowered epoch time to $\sim$ 40s for rapid debugging at the cost of final performance (71.30% test accuracy vs. 75.05% when fully fine-tuned). Overall, these system-level optimisations reduced the wall-clock time for a 10-epoch run from roughly 3.7 hours to about 12 minutes, enabling systematic experimentation with Triplet Loss on this backbone.
- **ResNet18 (PyTorch):** The PyTorch branch emphasised controlled architectural and augmentation studies rather than throughput engineering. Efficient data loading and on-GPU preprocessing kept training fast due to the smaller backbone and pairwise-loss formulation. We conducted two families of ablations: the **A-series**, varying augmentation strength (A0–A2), and the **R-series**, varying regularisation levels (R1–R3), which are referenced throughout the Results and Discussion.

Together, the two pipelines provide complementary perspectives: one focusing on system-level optimisation under Triplet Loss, and the other focusing on architectural and augmentation-level insights under Contrastive Loss.

4 Results

This section presents the quantitative and qualitative findings from both experimental tracks: the MobileNetV2 Siamese network trained with Triplet Loss and the ResNet18 Siamese network trained with Contrastive Loss. We report verification accuracy, ROC/AUC performance, confusion matrices at optimal thresholds, and qualitative analyses of embedding structure and failure cases. Together, these results provide a unified comparison of two lightweight metric-learning pipelines for face verification.

4.1 Evaluation Protocol

All results are computed on the identity-disjoint test split described in Section 3. For each model, we evaluate three components: (i) training dynamics, (ii) ROC/AUC performance (independent of threshold), and (iii) threshold-based verification accuracy. Optimal thresholds are chosen using the validation set and then applied to the test set. This ensures a fair and consistent comparison across models and loss functions.

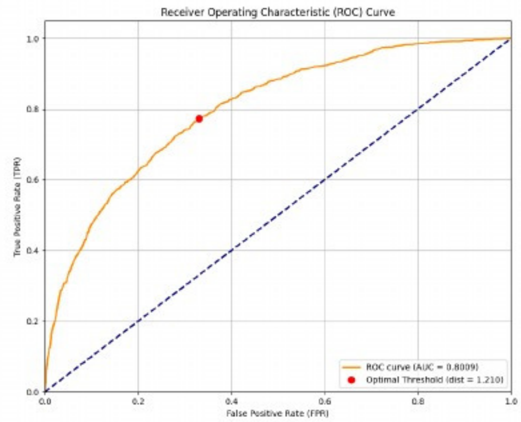
4.2 MobileNetV2 (Triplet Loss) — Summary Results

The final MobileNetV2 configuration (fully fine-tuned with mixed-precision training) achieves an AUC of **0.8009** and a verification accuracy of **72.12%** on the held-out test set. The corresponding optimal threshold is approximately **1.21**. The confusion matrix (Figure 2) shows balanced error patterns, correctly classifying 2012 negative pairs and 2315 positive pairs.

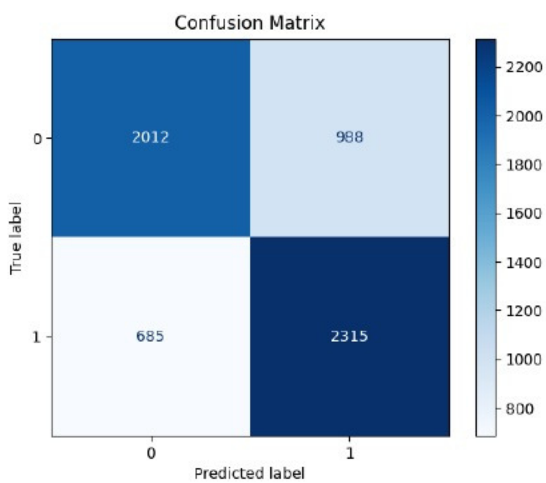
Across hyperparameter studies, three key findings emerged. First, increasing the triplet count from $N = 10{,}000$ to $N = 20{,}000$ degraded performance (75.05% to 73.60%), confirming that random triplet mining saturates quickly with easy negatives. Second, freezing the first 80 layers reduced training time ($\sim$40s per epoch) but lowered accuracy (71.30%), indicating that full fine-tuning is necessary to overcome the ImageNet–CelebA domain gap. Third, increasing the triplet margin from $\alpha = 0.5$ to 1.0 destabilised training, supporting the default margin as a local optimum for a 128-dimensional space.

These ablations were made possible by extensive system-level optimisation (Section 3), which reduced epoch time from $\sim$1313s on CPU to $\sim$70s on GPU. Figure 2 summarises the final model's training dynamics, ROC curve, and confusion matrix.

(a) Training and validation loss



(b) ROC curve (AUC = 0.8009)



(c) Confusion matrix at optimal threshold

Figure 2: Final MobileNetV2 results (Triplet Loss).

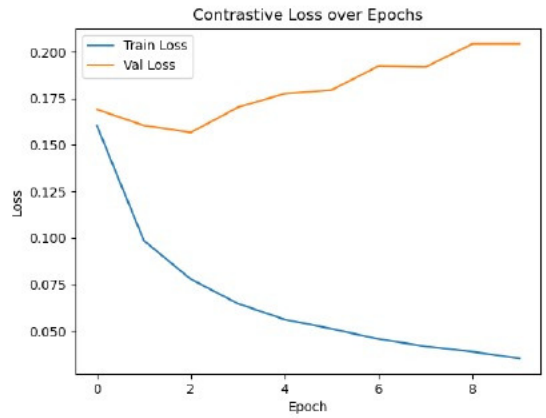
4.3 ResNet18 (Contrastive Loss) — Summary Results

The ResNet18 track produced a clear progression across architectural and augmentation choices. The baseline A0 model (frozen backbone, minimal augmentation) achieved an AUC of roughly 0.86 and an accuracy of 78%. Introducing stronger augmentation and unfreezing `layer4` (A2 configuration) improved the AUC to **0.9094** and accuracy to **82.95%**, with an optimal threshold of **0.57**.[2] The corresponding confusion matrix shows strong class separation: 1599 true negatives, 1719 true positives, and reduced false positives/negatives (401 and 281 respectively).

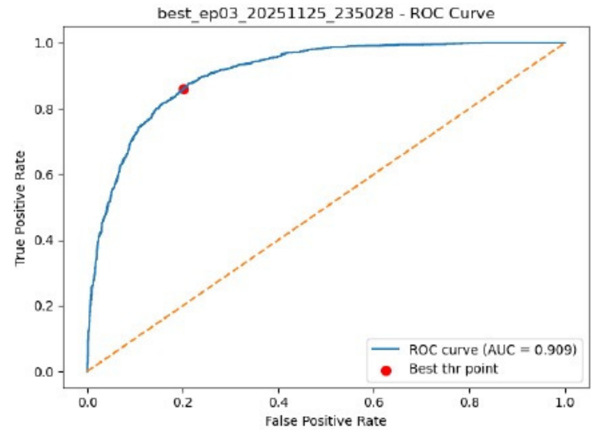
A regularised variant (R3) incorporating dropout (0.5) and weight decay (1e-4) produced more compact t-SNE clusters but slightly lower AUC (**0.8855**) and accuracy (80.65%), indicating that excessive regularisation can overly contract the embedding space.

Figure 3 visualises the A2 model's training dynamics, ROC curve, and confusion matrix.

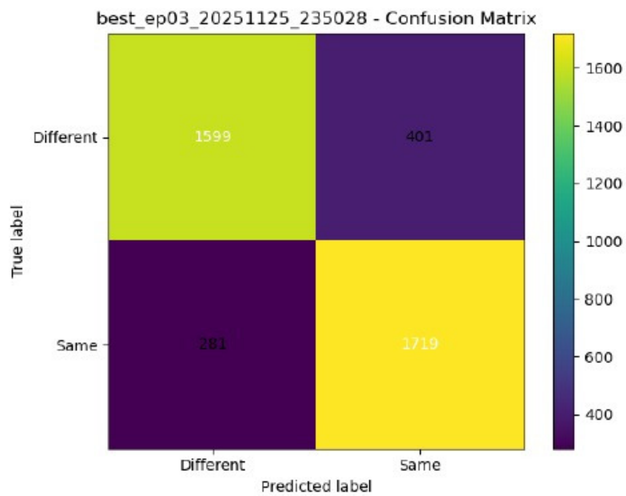
[2]ResNet18 A2 metrics: AUC = 0.909424, accuracy = 0.8295, threshold = 0.5700, TN = 1599, FP = 401, FN = 281, TP = 1719. See JSON file `metrics_extended_best_ep03_20251125_235028.json`.



(a) Training and validation loss



(b) ROC curve (AUC = 0.9094)



(c) Confusion matrix at optimal threshold

Figure 3: Final ResNet18 results (Contrastive Loss).

4.4 Comparative Summary

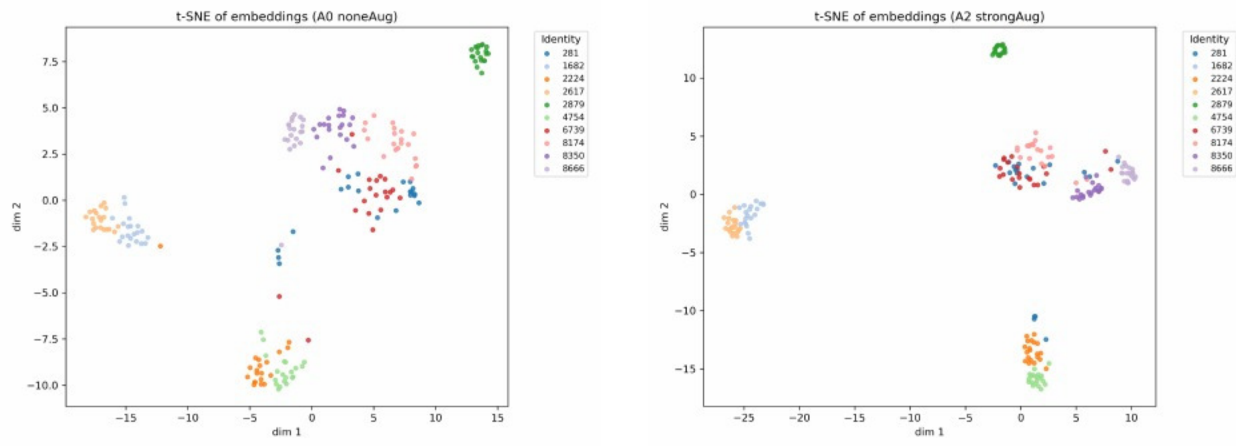
Table 1 summarises the key quantitative results across the two pipelines.

Model	Loss	AUC	Accuracy	Notes
MobileNetV2 (FT)	Triplet	0.8009	0.7212	Mixed precision best
ResNet18 A2	Contrastive	**0.9094**	**0.8295**	Final model
ResNet18 R3	Contrastive	0.8855	0.8065	More compact clusters

Table 1: Comparison of final models across both experimental tracks.

4.5 Qualitative Analysis

Embedding structure. Figure 4 compares t-SNE embeddings for A0 and A2. The A0 model produces overlapping identity clusters, while A2 forms well-separated, compact clusters, confirming the importance of augmentation and partial fine-tuning.



(a) A0 (no augmentation) (b) A2 (strong augmentation)

Figure 4: t-SNE embeddings showing improved identity separation from A0 to A2.

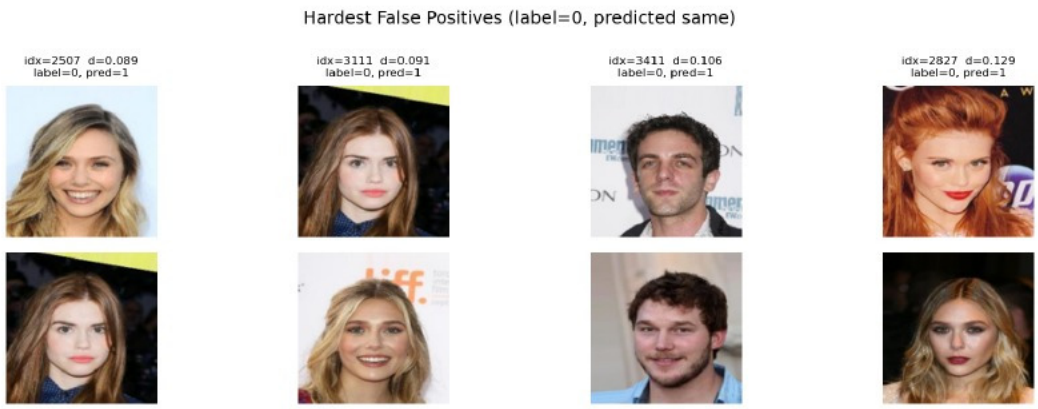


Figure 5: Hard false positives for the ResNet18 A2 model.

Failure mode analysis. Figures 5 and 6 show the hardest errors of the A2 model (at $\tau \approx 0.47$). The false positives have very small distances ($d \in [0.089, 0.129]$) and mostly confuse the same two female identities: both have pale skin, long wavy hair with a similar side part and a big smile, making them genuinely look-alike. The remaining column mixes two male identities with similar coarse face shape and hairstyle. In contrast, the hardest false negatives all correspond to one male identity pictured decades apart, with drastic changes in hair colour, hairstyle and facial shape; these pairs obtain large distances ($d \approx 1.04$–1.07), indicating that the model struggles to stay invariant to strong age- and hairstyle-related appearance changes.

Grad-CAM visualisation. Figure 7 shows Grad-CAM on two of the hardest false positives (top) and two false negatives (bottom). For the false positives, the network attends to central facial regions (eyes, nose, mouth and hairline), which look almost identical across the confused pairs; the model is therefore "wrong for the right reasons", collapsing genuinely similar faces. For the false negatives, attention shifts to features that differ strongly between photos (sunglasses and microphone versus unobstructed face; cap and forehead versus lower face), so the attended regions change with occlusions and ageing and the embeddings drift apart. This suggests brittleness mainly to occlusions and long-term appearance changes, rather than to moderate pose or illumination.

5 Discussion

Our study examined two complementary Siamese-network pipelines for face verification: MobileNetV2 with Triplet Loss and ResNet18 with Contrastive Loss. Although developed independently, the two tracks collectively illuminate how augmentation, fine-tuning, and embedding geometry influence verification performance in lightweight architectures.

Figure 6: Hard false negatives for the ResNet18 A2 model.

(a) False positive heatmap

(b) False negative heatmap

Figure 7: Grad-CAM analysis of misclassifications for the A2 model.

5.1 Interpretation of Results

A central theme across both branches is that **data quality and embedding structure** matter more than model size. The MobileNetV2 experiments show that a compact backbone can achieve reasonable performance (AUC 0.8009, accuracy 72.12%) when fully fine-tuned and supported by stable regularisation. Their ablations reveal well-known constraints of random triplet mining: early saturation, weak negative sampling, and sensitivity to backbone freezing.

The ResNet18 branch complements these findings by demonstrating the value of augmentation and selective fine-tuning. Strong augmentation and unfreezing `layer4` substantially improved performance (AUC 0.9094, accuracy 82.95%), with t-SNE visualisations confirming the formation of more coherent identity clusters. Qualitative analyses further clarify typical failure cases: false positives between visually similar identities and false negatives caused by occlusions or strong appearance changes. Grad-CAM heatmaps reinforce that errors often arise when attention shifts toward hair or background rather than identity-relevant regions.

5.2 Limitations

Both approaches face inherent limitations. For MobileNetV2, random triplet mining restricts exposure to challenging negatives, and the low input resolution limits fine-grained discrimination. For ResNet18, contrastive learning remains threshold-sensitive and struggles with extreme pose or illumination variation. A trial with cosine similarity performed poorly (AUC 0.62), suggesting that the current embedding space is better aligned with Euclidean distance. More broadly, pairwise metric learning provides only indirect supervision, which may leave subtle identity cues underrepresented.

5.3 Future Work

Future improvements should prioritise **online hard or semi-hard mining** to maintain informative gradients throughout training. Exploring angular-margin losses such as ArcFace or CosFace may yield stronger embedding geometry, while higher-resolution inputs or face-alignment preprocessing could mitigate common failure modes. Sample-balancing strategies and multimodal extensions (e.g., temporal cues or landmarks) also represent promising directions for improving verification robustness.

Overall, the combined insights from both branches highlight the importance of augmentation, fine-tuning, and data quality in shaping lightweight face-verification systems.

6 Conclusion

This work investigated two complementary Siamese-network approaches for face verification on CelebA, combining a MobileNetV2 backbone trained with Triplet Loss and a ResNet18 backbone trained with Contrastive Loss. Although differing in training objectives and optimisation strategies, both branches contributed meaningfully to understanding the design space of lightweight verification systems. The MobileNetV2 experiments highlighted the importance of domain adaptation and effective triplet selection, achieving stable performance following extensive pipeline optimisation. The ResNet18 branch extended these findings by demonstrating the strong impact of augmentation and partial fine-tuning, ultimately achieving higher discriminative capability with an AUC of 0.9094.

Beyond quantitative results, qualitative analyses using t-SNE, hard-pair inspection, and Grad-CAM provided insight into the structure of the learned embeddings and the factors underlying failure cases. These complementary perspectives reveal that robust verification performance emerges not from model size alone, but from thoughtful control of data diversity, embedding geometry, and fine-tuning behaviour.

Overall, the study underscores the viability of lightweight Siamese networks for face verification and identifies clear avenues for future improvement, including hard mining strategies, angular-margin losses, and higher-resolution inputs. Together, these findings offer a coherent foundation for continued exploration of efficient, interpretable metric-learning systems for identity verification.

References

Jane Bromley, Isabelle Guyon, Yann LeCun, Eduard Säckinger, and Roopak Shah. Signature verification using a "siamese" time delay neural network. In *Advances in Neural Information Processing Systems*, volume 6. Morgan Kaufmann, 1993.

Gregory Koch, Richard Zemel, and Ruslan Salakhutdinov. Siamese neural networks for one-shot image recognition. In *Proceedings of the ICML Deep Learning Workshop*, 2015.

Yaniv Taigman, Ming Yang, MarcÁurelio Ranzato, and Lior Wolf. Deepface: Closing the gap to human-level performance in face verification. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1701–1708. IEEE, 2014.

Florian Schroff, Dmitry Kalenichenko, and James Philbin. Facenet: A unified embedding for face recognition and clustering. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 815–823. IEEE, 2015.

Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4690–4699. IEEE, 2019.

Simone Bianco. Large age-gap face verification by feature injection in deep networks. *Pattern Recognition Letters*, 90: 36–42, 2017.

Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. Mobilenetv2: Inverted residuals and linear bottlenecks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520. IEEE, 2018.

Xiaoli Chen, Wei Zhang, and Li Wang. A lightweight face recognition model based on mobilefacenet for limited computation environment. *EAI Endorsed Transactions on Internet of Things*, 8(30):1–10, 2022.